

Information Congestion in AI Inference Routing: Market-Share Elasticity and Mechanism Design

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Abstract

Open-weight models create a real-time market for partly substitutable execution: harnesses originate token demand, routers buy execution, and providers sell perishable compute. We ask when price-sensitive routing makes adaptive pricing an information externality. If k providers respond to signals of effective rank $r_n \asymp n^\beta$, a simple congestion model has $k^* \asymp n^{(1+\gamma\beta)/(1+\gamma)}$: the efficient adaptive set is a vanishing fraction when $\beta < 1$ and a constant fraction when $\beta = 1$. Public quotes show a large mechanism surface but do not identify the regime. For GLM-5.2, three active undercutters in a 25-provider menu receive 18.62 percentage points of inverse-price shadow share from passive quotes; its rank slope is 0.967 (block-bootstrap interval [0.516, 1.219]), hence linear-compatible. Non-GLM active repricing is sparse. Paid rule interventions are promising but still accruing. A heterogeneous-provider game then shows that welfare-, revenue-, quality-, and viability-optimal router rules diverge. We derive score and capacity bounds and propose covariance-aware exposure caps. All empirical claims are property tests; none identifies collusion or welfare.

1 Introduction

Open weights separate a model from the firm that executes it. Two providers can serve the same model on the same prompt, making execution partly commodity-like. Yet tool calls, quantization, sampling, context limits, latency, privacy, and hidden capacity keep the products from being perfectly fungible. This produces a new vertical market: a user delegates to a harness; the harness chooses a model and possibly a router; the router aggregates token demand and dispatches an eligible provider; and inference providers sell perishable GPU-backed service.

The closest DeFi analogy is useful but incomplete. A harness resembles a wallet or intent originator, a router resembles a DEX aggregator, a model is a common execution standard, and a provider resembles a liquidity provider. Unlike an atomic swap, however, an inference request is normally assigned to a single provider, may be retried serially, has stochastic quality, and cannot be verified from price alone. The router is therefore not merely an aggregator. It is a delegated buyer, a multi-attribute scoring mechanism, and a reliability intermediary.

This distinction matters empirically. A public price is an invitation to route, not a firm capacity commitment. A documented score is not necessarily the allocation law after private eligibility, quality, and fallback. Provider cost depends on whether capacity is owned, reserved, or bought in a spot market; the same discount can reflect low marginal cost, subsidy, entry investment, or predatory intent. Consequently, price panels alone neither reveal flow nor identify dumping, collusion, or welfare.

We build the paper around this identification problem. Our evidence joins a five-minute public menu panel, router-published enforcement aggregates, model-author anchor prices, public quality fields, and randomized or assignment-first owned requests. The sources occupy different rungs of an identification ladder. Public menus reveal offers; public enforcement reveals router-side state; mechanical shadow routing reveals the implications of an explicit rule; owned requests reveal realized policy for one account and workload; and only router or provider logs could reveal market-wide flow or private capacity.

The empirical results are property tests and rule counterfactuals. On the immutable 22 July revision, the public panel contains 3.18 million endpoint observations and 3,274 snapshots. GLM-5.2 has three pre-holdout active undercutters in a median 25-provider menu. Resetting those quotes to the author benchmark transfers 18.62 percentage points of inverse-price shadow share back to passive providers. Yet the effective-rank ladder is linear-compatible at 1, 6, and 24 hour horizons, and a 70-market non-GLM census has zero active undercutters in 90% of markets. The desired GLM-minority/non-GLM-linear asymptotic contrast is therefore not an empirical finding. Paid routing gives a sharper intervention: in 15 complete blocks, explicitly requesting price sorting raises cheapest-provider selection by 80 points relative to default routing. The preregistered duration and choice gates have not passed.

We make five mechanism-design contributions.

- (1) We formulate separate user, router, provider, and planner objectives for inference routing. The formulation makes explicit why price-only, router-revenue, and welfare objectives need not agree.
- (2) We give a conditional information-congestion theorem. When provider signals have sublinear effective rank, the efficient number of adaptive pricers is sublinear even though each price cut can win share locally; with linear rank, the efficient set is a constant fraction.
- (3) We prove a score-price equivalence and elasticity decomposition. A provider's relative non-price score is a price-equivalent subsidy or penalty; its interaction with the price rule determines the return to undercutting.
- (4) For entropy-regularized routing, we prove an exact regret identity: using a price-only score instead of the full generalized execution cost loses exactly a scaled KL divergence. A uniform score-error bound then gives a robust approximation guarantee.
- (5) We prove a hidden-capacity impossibility result. Two economies with the same public menu can require opposite efficient allocations. Every quote-only router incurs at least half the cost gap in one of them. Execution-contingent capacity

certificates are therefore an informational requirement, not merely a feature request.

Finally, we evaluate the theorem and mechanism in two structural simulations. The fixed- n integer grid recovers the predicted scaling exponents to within 0.006, but a marginal-preserving bandit intervention changes action correlation without materially changing allocation. A heterogeneous-provider game recomputes bounded best responses under 108 router rules. It exposes a large conflict between router revenue and welfare and profitable pairwise joint deviations even when unilateral grid regret is zero. These are design screens, not live-market welfare or conduct estimates.

The paper contributes to multidimensional procurement [3], two-sided platforms [14], pricing technology [2], platform steering [9], and algorithmic interaction [6]. Its methodological position is deliberately conservative: when market-wide flow and cost are hidden, theory should state conditional guarantees and experiments should test mechanism properties before assigning conduct.

2 Market, Timing, and Objectives

2.1 Participants and transaction

A request has observable context x and user value $v(x, q)$ for delivered quality q . A harness transforms the task and chooses a model-router pair. The router observes a private eligibility set $A(x)$, public or private endpoint features, and possibly an ordered fallback list. Provider i posts input and output token prices, summarized by an expected request price $p_i(x)$, and delivers stochastic latency L_i , quality q_i , and success $y_i \in \{0, 1\}$.

Providers differ in capacity technology. An owned-capacity provider chooses K_i before demand, pays fixed cost $F_i(K_i)$, and may have low short-run marginal cost. A spot-dependent provider buys state-contingent capacity at $c_i(x, z)$ and has less fixed exposure. A model author can post a focal anchor price. A provider matching that anchor is “non-strategic” only with respect to its displayed-price process; it can still optimize admission, quality, or capacity.

Let s_i be the probability that i is attempted, including any position in a declared fallback policy, and let $d_i = s_i \mathbb{E}[y_i | i, x]$ be delivered share. Transfers cancel in static social surplus but affect entry and investment. The four objectives are

$$U_i = \mathbb{E}[(p_i - c_i(x, K_i, q_i))d_i - F_i(K_i) - \phi_i], \quad (1)$$

$$R = \mathbb{E}[f(p_i, d_i) - C_{\text{retry}} - C_{\text{SLA}}], \quad (2)$$

$$V = \mathbb{E}[v(x, q_i)y_i - p_i y_i - \lambda_L L_i - \lambda_F(1 - y_i)], \quad (3)$$

$$W = V + R + \sum_i U_i - \text{pure transfers}, \quad (4)$$

where ϕ_i includes penalties or collateral losses. A router may maximize revenue R , a user-facing generalized cost, or a constrained combination. We never infer which objective is deployed from public prices.

2.2 Information ladder

Table 1 states what each source can identify. The distinction between simulated share and realized share is essential. If an analyst applies $p_i^{-2}/\sum_j p_j^{-2}$ to a public menu, the output is a property of that formula and menu. It is not routed share unless eligibility,

Table 1: Identification ladder. Higher rows do not inherit the claims of lower rows without an explicit bridge.

Source	Identified object
Public menu	Quotes, endpoint labels, displayed availability
Router aggregates	Published congestion, derank, and rate-limit state
Shadow rule	Allocation implied by a declared formula, conditional on menu
Owned requests	Selected provider, cost, and outcomes for our account/workload
Private logs	Potentially arrivals, eligibility, attempts, and market-wide flow
Cost/value data	Needed with outcomes for welfare and dumping claims

quality filters, and fallback are either observed or experimentally shown irrelevant.

3 Conditional Mechanism Results

3.1 Inverse-power routing is a benchmark, not a measurement

Suppose, only for this subsection, that all n providers are eligible and the router really allocates

$$s_i(p) = \frac{p_i^{-a}}{\sum_j p_j^{-a}}, \quad a > 0. \quad (5)$$

With constant marginal cost c_i and fixed demand, provider profit is $\pi_i = (p_i - c_i)s_i$. This is a useful mechanism benchmark even though our data reject it as the full realized allocation law.

PROPOSITION 1 (CONDITIONAL SHARE ELASTICITY AND MARKUP). Under (5),

$$\frac{\partial \log s_i}{\partial \log p_i} = -a(1 - s_i). \quad (6)$$

Every interior best response satisfies the Lerner index $L_i = (p_i - c_i)/p_i = 1/[a(1 - s_i)] > 1/a$. If costs are symmetric, an interior symmetric equilibrium exists when $a(n - 1) > n$ and has

$$p^* = c \frac{a(n - 1)}{a(n - 1) - n}. \quad (7)$$

The result explains why the routing exponent controls price incentives *when the mechanism is actually used*. It does not license substituting $a = 2$ into a structural market model. Private eligibility and quality change both s_i and the derivative providers face.

3.2 Information congestion and the critical adaptive set

The elasticity result is individual: it says how one provider’s share changes with its own quote. It does not say how many providers should adapt to the same market signal. Let Σ_n be the covariance matrix of the information used by n potentially adaptive providers and define its effective rank

$$r_n = \frac{(\text{tr } \Sigma_n)^2}{\text{tr}(\Sigma_n^2)} \in [1, n]. \quad (8)$$

Low r_n means that many displayed responses are transformations of few common signals; it does not imply communication.

ASSUMPTION 1 (REDUCED-FORM INFORMATION CONGESTION). *If $k \in [0, n]$ providers adapt, their gross allocation benefit is bk/n and their social congestion loss is*

$$C_n(k) = c \left(\frac{k}{n} \right)^2 \left(\frac{k}{r_n} \right)^\gamma, \quad b, c, \gamma > 0. \quad (9)$$

The planner's reduced-form value is $V_n(k) = bk/n - C_n(k)$.

The first term is the local market-share opportunity. The second encodes a common-mode externality: correlated learners chase the same residual, overload the same capacity state, or repeatedly exploit the same scoring error. Its form is an assumption, not an implication of inverse-power routing.

THEOREM 1 (CRITICAL ADAPTIVE SET). *Under Assumption 1, the unique continuous interior optimum, before clipping to $[0, n]$, is*

$$k_n^\circ = \left(\frac{bnr_n^\gamma}{c(2+\gamma)} \right)^{1/(1+\gamma)}. \quad (10)$$

The integer optimum is one of $\lfloor k_n^\circ \rfloor$ and $\lceil k_n^\circ \rceil$. If $r_n \asymp n^\beta$ for $\beta \in [0, 1]$, then

$$k_n^* = \Theta\left(n^{(1+\gamma\beta)/(1+\gamma)}\right). \quad (11)$$

Consequently, $k_n^/n \rightarrow 0$ when $\beta < 1$, while $k_n^* = \Theta(n)$ when $\beta = 1$.*

The theorem separates two statements that are easy to conflate. A provider can have a large local return to undercutting while the efficient adaptive set is a vanishing fraction. Conversely, a common pricing response is not costly when it reflects linearly growing independent information. The sufficient statistic is signal rank relative to market size, not the count of low quotes.

COROLLARY 1 (ROBUST EXPOSURE CAP). *Let Ω be a declared uncertainty set for (b, c, γ, r_n) . The cap*

$$\bar{k}_n = \min_{\omega \in \Omega} [k_n^\circ(\omega)] \quad (12)$$

lies weakly below the increasing portion of $V_n(k; \omega)$ for every $\omega \in \Omega$. It prevents overexposure under all declared models, at the cost of possibly excluding beneficial adaptation.

Operationally, the cap is a bound on correlated exposure, not on provider entry. A router can preserve exploration and redundancy while limiting the aggregate attempt probability assigned to providers whose price innovations share a factor. The empirical burden is substantial: a cap is justified only after effective rank transports to a holdout and the congestion loss is calibrated with realized outcomes.

3.3 Displayed price and latent score

Let α_i denote a stable reduced-form non-price score on a fixed request class and exact eligible menu. The score-adjusted rule is

$$s_i(p, \alpha) = \frac{p_i^{-a} e^{\alpha_i}}{\sum_j p_j^{-a} e^{\alpha_j}}. \quad (13)$$

The parameter is deliberately not called quality. It bundles quality, latency, health, capacity, eligibility mismatch, and any persistent router preference not represented by public price.

PROPOSITION 2 (SCORE-PRICE EQUIVALENCE AND ELASTICITY). *Under (13), define $p_i^{\text{eff}} = p_i \exp(-\alpha_i/a)$. Then $s_i(p, \alpha) = s_i(p^{\text{eff}}, 0)$ and, for any two providers with positive conditional selection probability,*

$$\alpha_i - \alpha_j = \log \frac{s_i}{s_j} + a \log \frac{p_i}{p_j}. \quad (14)$$

If $\alpha_i = h_i(\log p_i)$ while rival scores and prices are locally fixed,

$$-\frac{\partial \log s_i}{\partial \log p_i} = (1 - s_i)\{a - h'_i(\log p_i)\}. \quad (15)$$

Equation (15) is the mechanism object. A score that falls with an aggressive cut has $h'_i > 0$ and attenuates price-induced share capture; a score that rewards the same cut beyond its explicit price term has $h'_i < 0$ and amplifies it. Equation (14) identifies a relative level wedge from repeated owned choices conditional on exact menus and a frozen a . It does not identify the derivative without score movement or a rule intervention.

For a below-benchmark provider, let $p^{(b,i)}$ replace only p_i by the model-author benchmark. Define the unilateral price-induced share gain and its score interaction as

$$\Delta_i(\alpha) = s_i(p, \alpha) - s_i(p^{(b,i)}, \alpha), \quad (16)$$

$$\Gamma_i = \Delta_i(\alpha) - \Delta_i(0). \quad (17)$$

Negative Γ_i means scoring attenuates the share benefit of undercutting; positive Γ_i means it amplifies it. These one-provider counterfactuals are not additive when several providers undercut simultaneously.

3.4 The cost of omitting quality

Let the generalized execution cost of provider i be

$$g_i = a \log p_i + \lambda_L \widehat{L}_i + \lambda_F \widehat{r}_i - \lambda_Q \widehat{q}_i - \lambda_K \log \widehat{K}_i, \quad (18)$$

where $\widehat{r}_i$ is failure risk and $\widehat{K}_i$ is committed, not merely advertised, capacity. For x in the probability simplex Δ_n , define the entropy-regularized objective

$$F_g(x) = \langle x, g \rangle + \eta^{-1} \sum_i x_i \log x_i, \quad \eta > 0. \quad (19)$$

Its minimizer is $x_i^g \propto e^{-\eta g_i}$. A price-only router instead uses $\bar{g}_i = a \log p_i$ and $x^{\bar{g}}$.

THEOREM 2 (EXACT OMITTED-ATTRIBUTE REGRET). *For any finite score vector g and any allocation $x \in \Delta_n$,*

$$F_g(x) - F_g(x^g) = \eta^{-1} \text{KL}(x \parallel x^g). \quad (20)$$

In particular, the loss from price-only routing is exactly $\eta^{-1} \text{KL}(x^{\bar{g}} \parallel x^g)$ and is zero if and only if the two allocations coincide. Omitted attributes that are constant across eligible providers do no harm; attributes that reorder providers generally do.

The identity is sharper than saying quality “may matter.” It turns the distance between a price-only and full score into an exact objective gap for a declared regularized objective. It also gives a measurable target: a router can publish both score components and report the implied KL gap without disclosing prompt contents.

THEOREM 3 (ROBUST APPROXIMATE SCORING). *Suppose $\|\widehat{g} - g\|_\infty \leq \epsilon$ and let $x^{\widehat{g}}$ minimize $F_{\widehat{g}}$. Then*

$$0 \leq F_g(x^{\widehat{g}}) - F_g(x^g) \leq 2\epsilon. \quad (21)$$

Thus an auditable score can be approximately efficient without perfectly estimating every attribute. The guarantee is pointwise and does not assume a provider behavioral model. It does require that the score target g encode the planner or user objective rather than an ad hoc proxy.

3.5 Why public prices cannot recover hidden capacity

THEOREM 4 (QUOTE-ONLY HIDDEN-CAPACITY IMPOSSIBILITY). *Consider two eligible providers with identical observable quotes, public quality, and public reliability. There exist two economies consistent with those observables: in economy A, provider 1 has committed capacity and social cost c while provider 2 has scarcity cost $c + \Delta$; in economy B the costs are reversed. Any quote-only randomized router incurs expected cost regret at least $\Delta/2$ in one economy. A deterministic quote-only router incurs regret Δ in one economy.*

The theorem formalizes the reserved-capacity problem. Fundraising, data-center ownership, or a provider name is not the missing statistic: those are noisy proxies for deliverable capacity. The mechanism needs a commitment whose breach has a consequence. A capacity certificate can be a finite-slot report backed by collateral, an execution-contingent price, or an SLA with measurable credits.

3.6 Fee conflict

Suppose the router receives an ad-valorem fee τ on selected spend. Holding volume and allocation fixed, its revenue is $\tau \sum_i x_i p_i$, while the user’s payment is $\sum_i x_i p_i$. A common price increase therefore benefits the router and harms the user one-for-one up to τ . A flat fee per successful request is locally neutral to the selected price but can still distort admission or volume. Fee separation does not solve mechanism design, but it removes the most direct price-level conflict.

4 Data and Property Tests

4.1 Capture system and estimands

The main public panel begins on 7 July 2026 and, in the frozen release used here, runs through 22 July. It captures endpoint and model state at five-minute cadence in a remote GitHub Actions–Hugging Face pipeline. The immutable revision `f4023bdad64b6ad468741a5a3b1f3afb7af40dd0` contains 3,182,780 endpoint observations, 3,274 snapshots, 321 model identifiers, and 72 provider labels. We construct completion-price events only when a same endpoint-model quote changes between valid snapshots. We separately collect router-published rate-limit, derank, and capacity fields. Assignment-first paid probes record treatment before sending a minimal request and retrieve the selected provider and outcome for our account. Private prompts and request rows never enter public releases.

Every result freezes a dataset revision and declares a calibration/holdout split. The information-congestion audit trains through 16 July 13:52 UTC and evaluates later snapshots. A clean integrity rerun executes 29 registered public modules with zero failures and explicitly excludes prospective owned outcomes. These windows

are short; they support falsification and mechanism screening, not a mature structural estimate.

The primary price and competition population is paid-only. We exclude model IDs ending in `:free`, the `openrouter/free` meta-route, and non-positive completion quotes; paid suffixes such as `speed-` or `price-priority` routes are not excluded merely because their IDs contain a colon. Free-tier flow remains a separate subsidy/demand-creation estimand. In this frozen revision, 14 of 338 catalog entries are free-labelled and free routes account for 10.9% of tokens on the latest complete activity day, but zero of the 766 positive completion-price events. Thus the paid filter changes demand totals, not the focal pricing-event results. Source-defined “Other” aggregates and app-wide rankings lack variant identity and are not claimed to be cleanly filterable.

4.2 Regimes are persistent, except active undercutting

Provider-model pairs are frozen in a training window into four descriptive pricing regimes relative to the model-author benchmark: premium, anchor adopter, static discounter, and active undercutter. The names describe a price path, not a provider’s cost or intent. Of 366 training pairs, 358 are holdout-evaluable. Overall persistence is 0.944 (95% Wilson interval $[0.915, 0.964]$). Premium, static-discount, and anchor regimes persist at 0.992, 0.976, and 1.000, respectively; active undercutting persists in only 5 of 21 pairs, or 0.238 ($[0.106, 0.451]$). Fourteen of the 21 active pairs become static discounters.

This pattern fits a Brown–MacKay-style world in which repricing technologies differ [2], but it does not identify such technology. Active undercutting could also reflect promotion windows, capacity shocks, or menu corrections. The testable implication is duration: future data should estimate transition hazards as a function of benchmark gap, capacity state, and update frequency, with provider-model fixed effects.

4.3 The displayed-price transfer surface

We evaluate a 1,000-input-token, 256-output-token request at all 3,274 snapshots. A provider is “active” only if it changes price at least twice before the temporal split and its median quote lies below the author/menu benchmark. The classifier selects Novita, StreamLake, and Inceptron for GLM-5.2: three providers in a median menu of 25. For each post-split snapshot, we reset a nested active set to the benchmark and recompute the public inverse-price surface.

The one-, two-, and three-provider sets receive 11.36%, 22.81%, and 27.61% of shadow share. Their benchmark-reset counterfactuals are 2.64%, 5.82%, and 8.99%, so the implied transfers from passive providers are 8.72, 16.99, and 18.62 percentage points. These are exact rule effects conditional on the public menu. They are not realized flow or a demand estimate.

For GLM-5.2 the fitted effective-rank exponent is 0.967 with block-bootstrap interval $[0.516, 1.219]$ at one hour. The six- and 24-hour estimates are 1.098 $[0.714, 1.317]$ and 1.121 $[0.809, 1.326]$. The requested $k^* = o(n)$ reading therefore does not follow. Across 70 non-GLM paid markets, the cross-market active-count exponent is 0.219 $[0.068, 0.378]$ and 90% have zero active providers. The only supported non-GLM rank ladder has an in-sample common factor

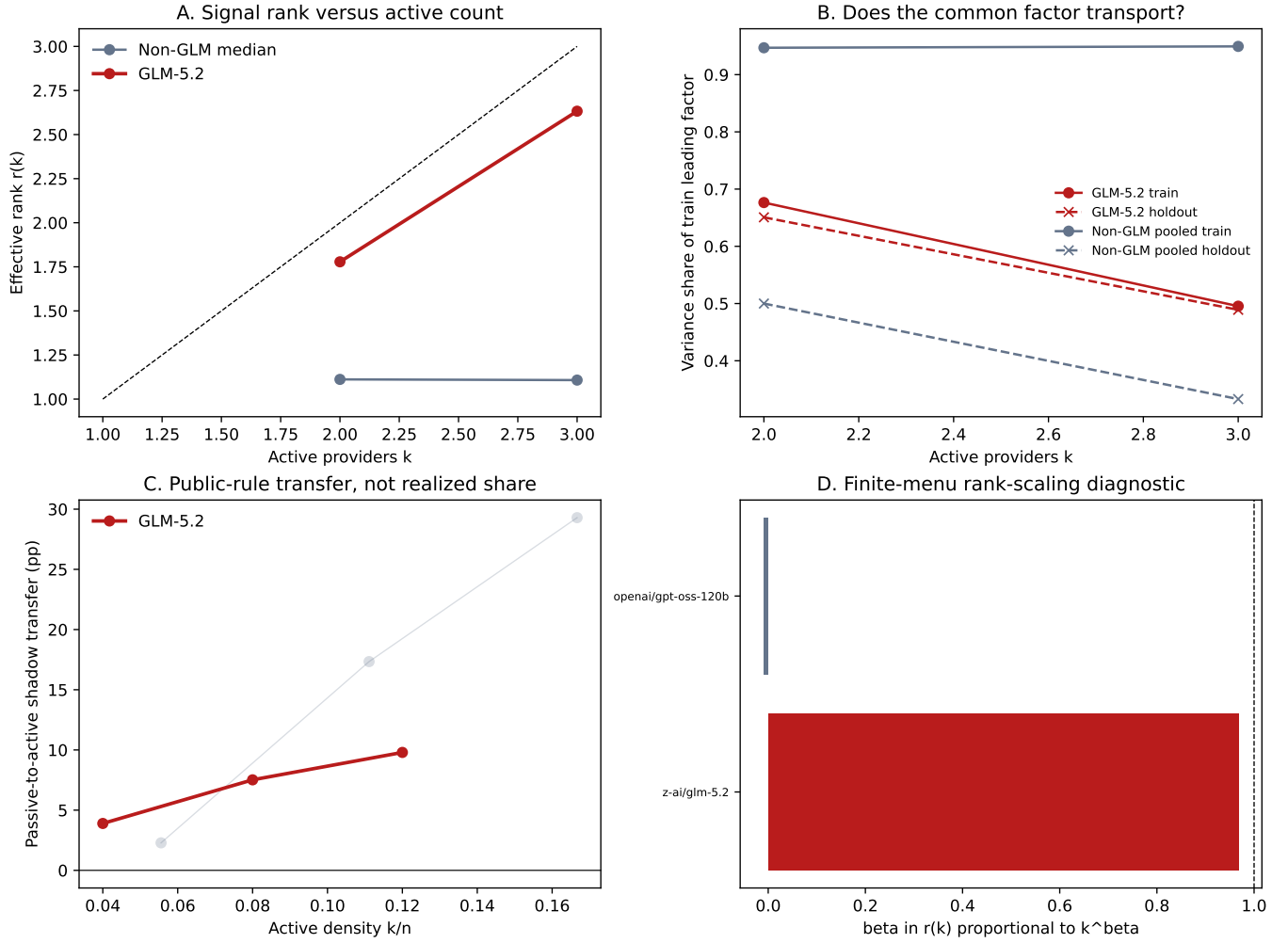


Figure 1: Finite-menu information-congestion property test. GLM-5.2 has a large public-rule transfer but a linear-compatible rank slope. The non-GLM low-rank fit does not transport to its holdout. Slopes are not asymptotic estimates and shadow allocations are not realized market shares.

that collapses in the temporal holdout. Current non-GLM data are sparse, not evidence for $k = \Omega(n)$.

4.4 Prospective identification of scoring beyond price

The GLM-5.2 paid campaign freezes two default-route requests, one explicit price-sort request, pins, and exclusion controls in each 15-minute block. A prospective amendment begins at 22:00 UTC on 21 July, after four earlier default choices were observed; those four choices are excluded from this analysis. Conditional multinomial choice estimates (13) with $a = 1.6483$ and a fixed ridge penalty. It reports relative α_i versus Z.AI, the price-equivalent discount $1 - e^{-\alpha_i/a}$, whole-block cluster intervals, and the mean within-menu total variation between price-only and score-adjusted probabilities. Five-fold whole-block log-loss improvement measures information

beyond price in bits per choice; a conditional Monte Carlo test holds every menu and quote fixed.

The interaction in (17) is the primary economic output. The within-block price-sort arm separately estimates how explicitly changing the routing rule affects cheapest-provider selection. Diagnostic estimates require 40 covered prospective choices, 20 blocks, and three selected providers; paper-strength interpretation requires 800 choices, 100 blocks, seven days, and 90% candidate coverage. The current aggregate has 36 covered default choices in 18 blocks over 0.647 days. Its public-price exponent is 1.735 with profile interval $[0.31, 3.16]$. The focal active pair is selected 2/36 times, versus a 13.60% frozen-rule prediction; the observed 5.56% share has Wilson interval $[1.54\%, 18.14\%]$. The score model has only 30 prospective choices in 15 blocks and is not fit.

The direct rule contrast is sharper. In 15 complete blocks, default routing selects the cheapest provider 0% of the time and explicit price sorting does so 80% of the time; the block-bootstrap interval

for the 80-point difference is [60, 100]. This owned-request effect shows that rule choice matters and that default routing is not price-only. It does not reveal the proprietary score or market-wide flow, and remains accruing under the registered gate.

4.5 No detectable excess rival response in the active regime

For 1,052 active-undercutter events in 11 target provider-model clusters, the registered rival-action rate is 0.455 ([0.425, 0.486]), compared with 0.453 ([0.424, 0.484]) in a shifted timing placebo. The paired difference is 0.0019; a target-cluster bootstrap interval is [−0.0438, 0.0410], and the exact one-sided test for *more* registered-window response has $p = 0.468$. The placebo is not randomized treatment. The result places the immediate response mechanism near zero at this cadence but does not exclude slower, state-dependent, or provider-specific response.

4.6 Displayed score is not realized clearing

An independent welfare audit compares the elasticity implied by the public inverse-square score with the observed aggregate share elasticity. The inverse-square restriction is rejected with $z = 10.06$. An earlier aggregate estimate is -1.19 (standard error 0.17), but it is neither request-level selection nor a price experiment. We therefore use inverse-power routing only as the conditional benchmark in Section 3.

Owned routing reinforces the distinction. In the provider-regime release, 173 delegated selections match frozen regimes: 81 anchor and 92 premium, with no current delegated support for the active or static regimes. These counts depend on our model and prompt mix and are not market shares. A separate frozen first-position experiment, H81, decomposes fallback from delegated selection. In its 133 preterminal assignments, the missing-outcome identified set places the fallback and total-delegation contrast at 4.4–6.7 percentage points and hidden selection at zero. Missingness suppresses the complete-data primary randomization family, and two repeated models prevent broad transport. The fixed-horizon H95 replication remains blinded until its preregistered horizon and is not pooled with H81.

4.7 Unit share elasticity is not a revenue first-order condition

The pooled public quantity-share regression gives a price coefficient of -1.104 (clustered standard error 0.079), statistically equivalent to -1 within the registered 0.25 margin. That resemblance is not a residual-demand derivative. The within-provider/model coefficient is -0.605 (0.196), while the between coefficient is -1.184 (0.195); the between-minus-within gap is -0.579 with $p = 0.0031$. Only 68 listed-price events are available, 10.5% of listed entities vary price, and within-price variation fails the promotion gate.

There is also an exact accounting explanation. Let s_i be token share, $e_i = p_i q_i / \sum_j p_j q_j$ be public price-times-token revenue share, and $P = \sum_j p_j q_j / \sum_j q_j$. Within a model-day market,

$$\log s_i = \log e_i - \log p_i + \log P. \quad (22)$$

The revenue-share price coefficient is -0.104 (0.079), so the quantity-share coefficient is exactly one lower. The identity holds to $3.6 \times$

10^{-15} . It explains why a pooled slope near -1 can coexist with price-neutral revenue shares; it does not explain endogenous price-quantity matching or identify profit maximization.

A bounded unilateral counterfactual places the observed proxy-weighted provider revenue gap at 1.87–1.97% under the registered demand scenarios, with a combined bootstrap and elasticity envelope of 1.65–2.03%. This is conditional on treating the public elasticity as causal and extending it with a logit curve. Router revenue is not point identified because the applicable take rate, end-user price, and market demand curve are unobserved.

4.8 Quality, capacity, and “dumping” remain unidentified

Premium providers do not show a significant holdout throughput advantage ($p = 0.507$) or latency advantage ($p = 0.796$) in provider-model residual aggregates. This does not prove premiums lack value: task fidelity, privacy, support, and selection into quality measurement remain missing. More sharply, zero provider-model cells have a valid utilization measure linking load to a capacity ceiling. Static discounters more often fall below one scenario GPU- cost bound, but marginal cost, reserved-capacity amortization, rebates, and subsidies are unobserved. The registered verdict is therefore `dumping_supported=false`.

The same caution applies to geography. A name-based “China-linked” tag is more common among active and static discount regimes, but a name tag is neither cost nor strategy. A causal explanation requires at least location, hardware, energy price, reserved capacity, subsidy/rebate exposure, and realized utilization. These are fields for a data partnership, not coefficients to impute from names.

4.9 Dynamic score memory is outside the identified mechanism

The estimand in this paper is contemporaneous: conditional on the current menu, how much does displayed price move the declared allocation rule, and how much does the current reduced-form score attenuate or amplify that movement? Neither the public panel nor the owned-choice design identifies a law of motion for the score. In particular, persistence of a provider’s quote is not evidence that a router remembers the quote, and a lagged provider response is not evidence that the router carries a hidden penalty state.

We therefore remove critical-memory and signal-coupling simulations from this paper’s evidentiary chain. They remain separate learning benchmarks. A dynamic score model can enter this mechanism paper only after lagged price and quality histories improve held-out prediction of owned routing choices beyond the current menu, provider-model effects, enforcement state, and contemporaneous quality, and after a randomized router or partner-log design distinguishes score memory from unobserved eligibility. Until then, α_{it} is a current composite residual, not a state variable with an estimated transition.

5 An Auditable Router Design

5.1 Score, commitments, and exploration

Theorems 2–4 suggest a concrete design.

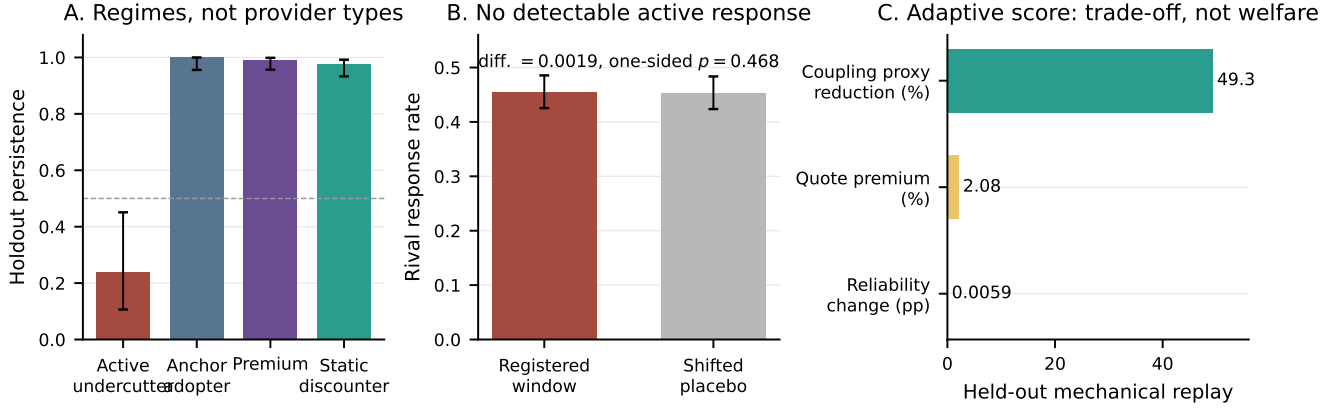


Figure 2: Three empirical boundaries. Panel A: frozen training regimes and holdout persistence (Wilson 95% intervals). Panel B: active-undercutter rival response in the registered window versus a shifted timing placebo. Panel C: held-out adaptive-score replay; the reliability change is in percentage points. Panel C holds menus and provider behavior fixed, so it is not a welfare estimate.

- (1) Compute a generalized score like (18), with task-conditional latency, failure, and quality predictions estimated out of sample.
- (2) Replace advertised capacity with an execution-contingent commitment: finite slots, collateral, or SLA credits. Publish how retries enter score and payment.
- (3) Use bounded exploration so new or small providers can generate evidence; cap concentration and common-mode exposure.
- (4) Charge the router through a fee separated from selected spend, while reporting both user generalized cost and router revenue.

The score need not collapse every objective into one undocumented number. The router can publish a Pareto frontier over user generalized cost, router revenue, provider viability, concentration, and completion. Different user classes can choose points on that frontier, provided the accounting is explicit.

5.2 Three objective vectors, three mechanisms

The same evidence can support several explicit objectives without pretending they coincide. Let x be delivered allocation and let q_i, r_i, L_i, c_i denote verified quality, failure risk, latency, and resource cost:

$$G_U(x) = \sum_i x_i [p_i + \lambda_L L_i + \lambda_F r_i - v(q_i)], \quad (23)$$

$$G_W(x) = \sum_i x_i [c_i + \lambda_L L_i + \lambda_F r_i - v(q_i)], \quad (24)$$

$$G_R(x) = - \sum_i x_i [f_i(p_i) - C_{\text{retry},i} - C_{\text{SLA},i}]. \quad (25)$$

An entropy-regularized router can minimize any one by using its corresponding score. The user-cost and welfare rules differ because price is a transfer in G_W but a cost in G_U ; an ad-valorem revenue rule may rank a high price above a low price; and a quality-first rule can rationally buy expensive execution when verified fidelity is valuable. None dominates without value, cost, and participation restrictions.

Our implementation recommendation is a declared Pareto menu, not one secret scalar: (i) a user generalized-cost arm, (ii) a resilience-constrained welfare arm using cost bounds and capacity commitments, and (iii) a quality-priority arm with a spend cap. Router revenue is reported as an outcome or constrained floor, not silently inserted into the user score. Provider entry, investment, and best responses are recomputed for each arm in simulation; the live experiment randomizes only arms that pass completion and fidelity safety floors.

5.3 Robust policy optimization

Let θ parameterize score coefficients, exploration, caps, and fee design. Let Ω be a declared set of cost, quality-value, capacity, and provider-learning models consistent with the evidence. One implementable objective is

$$\min_{\theta} \sup_{\omega \in \Omega} \mathcal{R}_U(\theta; \omega) + \lambda_P \max_i \mathcal{E}_i(\theta; \omega) + \lambda_C \max_{S: |S| \leq k} \mathcal{E}_S(\theta; \omega) + \lambda_V \mathcal{V}(\theta; \omega), \quad (26)$$

subject to completion, concentration, budget, and capacity constraints. $\mathcal{R}_U$ is user regret relative to the best feasible full-information policy; $\mathcal{E}_i$ and $\mathcal{E}_S$ are unilateral and coalition exploitability; and $\mathcal{V}$ collects constraint violations. This is not a claim that the router knows welfare. It is a way to report which uncertainties drive a design recommendation.

5.4 Structural equilibrium and objective frontier

We construct ten declared provider technologies: an author, two anchor resellers, two providers with reserved capacity, three spot-dependent startups, and two custom-quality providers. Technologies differ in marginal and fixed cost, capacity, reliability, latency, and quality. Anchors hold the benchmark price; the other providers play cyclic bounded price best responses. We screen 108 router rules

over price exponent, quality and reliability weights, uniform exploration, and a cap on providers assigned to one correlated-pricer group. Payments cancel in welfare. Every selected rule reports unilateral grid regret and a same-type two-provider joint-deviation screen.

The inverse-square price-only baseline has welfare 137.29, router fee revenue 5.99, and 111.52 quality-adjusted successes in simulation units. The welfare- selected rule raises welfare to 208.44 and quality-adjusted successes to 149.96, but lowers router revenue to 4.88 and leaves eight providers viable. The router-revenue rule raises revenue to 7.78 while lowering welfare to 83.99 and user utility below zero. The selected rules have zero unilateral regret on the registered grid, yet the largest bounded pairwise joint deviation gains 1.87 under the welfare rule and 8.88 under the revenue rule. Technology shocks to reserved capacity, reliability, spot cost, and quality are reported separately. The exercise shows objective conflict and coalition susceptibility, not an estimated equilibrium or evidence of collusion.

5.5 The policy-frontier experiment

The structural frontier suggests a prospective experiment whose estimands are narrower than “welfare” but decision-relevant. Randomize request triplets within model-prompt-time blocks among (i) the production delegated policy, (ii) a price-ordered policy with fallback, and (iii) the adaptive generalized score. Freeze the adaptive score before opening outcomes. The primary vector is attempted-request spend, completion, total latency including retries, and a blinded task-fidelity score. Provider choice, fallback position, and public enforcement state are mechanism variables rather than outcomes to condition on.

Because routing treatments can change congestion, inference should randomize time clusters or use switchbacks after a low-load pilot, report assignment-first intention-to-treat effects, and preserve bounded-outcome randomization intervals. No scalar index should be primary unless the user-value weights are frozen in advance. Instead, report the joint Pareto set and the range of value weights for which each policy is preferred. A second, provider-facing phase randomizes execution-contingent capacity commitments among willing endpoints and measures admission, delivery, and post-experiment retention. This separates the short-run routing effect from the longer-run entry and investment response that a menu replay cannot recover.

The experiment has three promotion gates. First, the adaptive arm must preserve randomization integrity and overlap across models, hours, and provider sets. Second, its completion and fidelity lower bounds must exceed declared safety floors. Third, any welfare statement must remain valid over a preregistered box of user values and provider cost bounds. Failure of the third gate still permits a useful multi-outcome mechanism comparison; it only blocks scalar welfare.

5.6 Practical provider types

The mechanism must work across three economic technologies even though public price regimes do not identify them.

- *Owned or reserved capacity*: high fixed cost, low short-run marginal cost, and a strong incentive to maintain utilization. Capacity

commitments are credible when backed by sunk infrastructure but still require outage penalties.

- *Spot-dependent providers*: flexible entry and state-dependent cost. They provide resilience but may disappear exactly during common scarcity. State-contingent commitment prices are more appropriate than a fixed share.
- *Model authors and anchor adopters*: an author quote can coordinate expectations, while identical third-party quotes may reflect default menus rather than common costs. The router should not treat an anchor as a cost certificate.

Free entry is not automatically efficient. An entrant adds a redundant delivery path but diverts requests and may increase retry or monitoring costs, echoing the classic business-stealing result [11]. Capacity-certified entry is preferable to counting endpoint labels: it rewards incremental delivered capacity rather than duplicated menus.

6 Related Work

Multi-model routers such as FrugalGPT, RouteLLM, and RouterBench optimize across models [4, 8, 13]. We hold the model label fixed and study provider procurement; the two routing layers compose. Multidimensional procurement shows why price and quality must be scored jointly [3]; two-sided platform theory clarifies fee and participation externalities [14].

Brown and MacKay study heterogeneous pricing technology and repricing frequency [2]. Their framework motivates our duration and regime transition tests, not a provider-type label. Hansen, Misra, and Pai show that independent, misspecified pricing learners can produce supra-competitive prices through correlated experiments [6]; our ordered property chain and signal-order intervention isolate that mechanism and currently fail to transport it. Johnson, Rhodes, and Wildenbeest show how platform prominence can change algorithmic competition [9]. Inference routing adds hidden eligibility, fallback, and stochastic delivered quality.

The closest empirical benchmark uses OpenRouter and Azure data to study the emerging market for intelligence [5]. We focus on provider-level microstructure and the gap between displayed and delivered liquidity. Switchback and marketplace experiments face interference [1, 7]; our assignment-first probes and first-position estimands are designed around that problem. Price focal points and the reflection problem motivate hard nulls for anchor following rather than causal stories from ties alone [10, 12].

7 Limitations and Conclusion

The public panel is measured in weeks, not quarters. Owned probes cover one account, region, workload, and budget. Capacity, provider cost, user task value, private rebates, and market-wide flow are not observed. Public enforcement fields are router-published aggregates rather than request logs. The current quality panel is selected and incomplete. Public shadow rules hold eligibility fixed; simulator interventions are causal only inside the simulator. We therefore do not estimate dumping, literal front-running, communication, collusion, or a scalar welfare loss.

Within those limits, the market description changes sharply. Open-weight execution is not a transparent commodity exchange. It is a two-layer procurement market in which public prices sit above

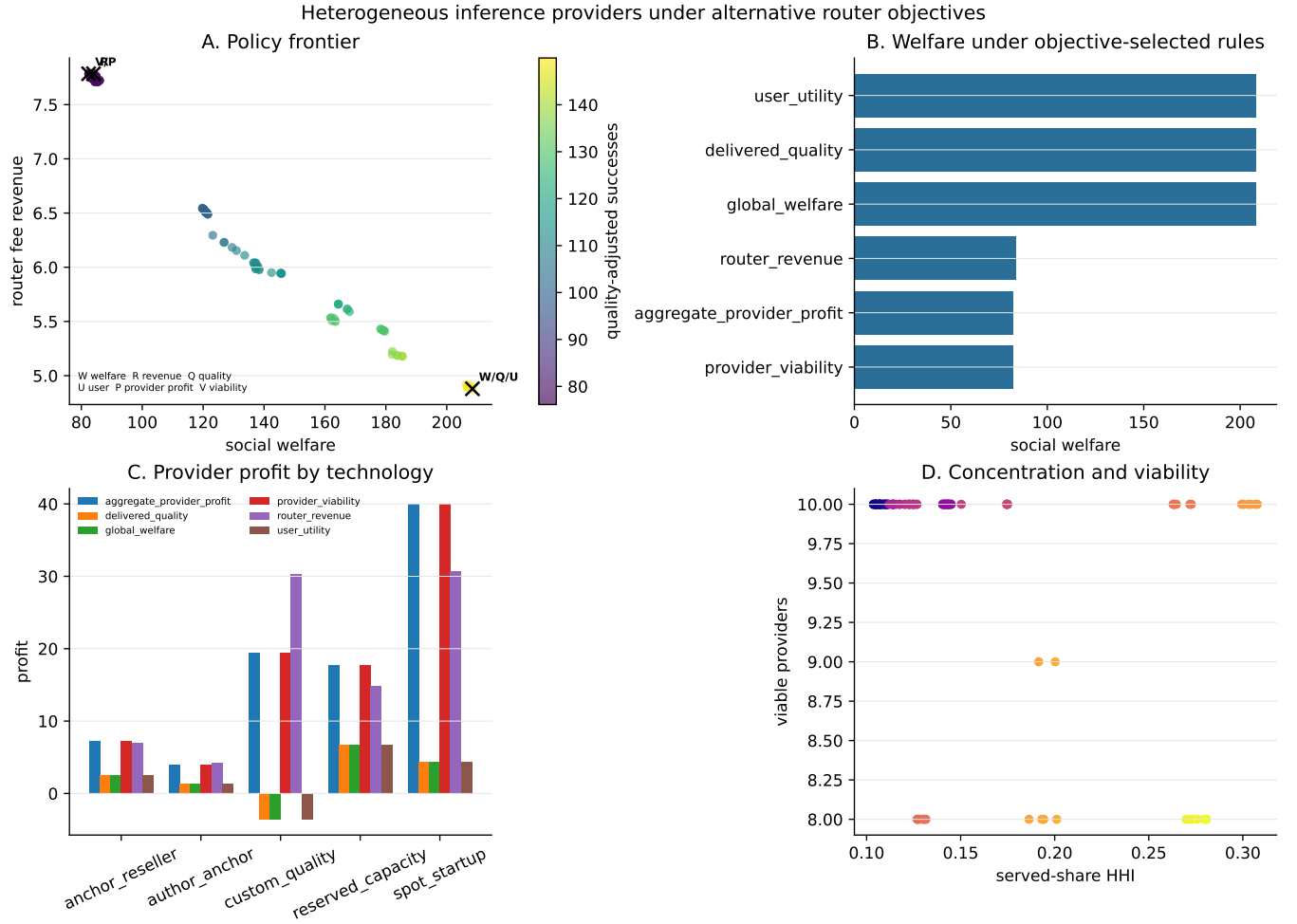


Figure 3: Structural router-objective screen. The price-only inverse-square baseline is marked against rules selected for welfare, router revenue, delivered quality, provider viability, user utility, and aggregate provider profit. Technology and demand are scenarios; the figure is not a live-market estimate.

hidden admission, scoring, and fallback. The empirical evidence rejects several tempting simplifications: inverse-square simulated share is not realized flow; active undercutting is not a stable provider type; current response timing does not support a simple reactive collusion story; pooled unit share elasticity is not an own-price first-order condition; and premium prices are not yet explained by observed QoS. It also refuses the paper’s most tempting positive contrast: current GLM rank is linear-compatible and non-GLM active repricing is sparse, so neither proposed asymptotic regime is empirically established.

The design response is not to declare a better exponent. It is to expose the missing contracting and information dimensions. The critical-set theorem says when correlated adaptation can require a minority exposure cap; the data say that premise is not yet established. Generalized scoring has an exact regret interpretation, approximate scores have a robust guarantee, and hidden capacity creates a formal impossibility for quote-only routing. Execution-contingent prices, covariance-aware caps, verified quality, bounded

exploration, and fee separation turn those results into testable mechanisms. The next empirical step is a randomized policy frontier with delivered outcomes and declared value bands. Until then, the honest output is a set of feasible mechanisms, not one estimated welfare optimum.

A Proofs

Conditional share elasticity and markup. Write $W_{-i} = \sum_{j \neq i} p_j^{-a}$. Direct differentiation gives

$$\frac{\partial s_i}{\partial p_i} = -\frac{a}{p_i} s_i (1 - s_i).$$

The first-order condition for $(p_i - c_i)s_i$ is $1 - a(p_i - c_i)(1 - s_i)/p_i = 0$, yielding the Lerner expression. At a symmetric profile $s_i = 1/n$; solving for p gives the formula. A finite positive solution requires $a(n - 1) > n$. $\square$

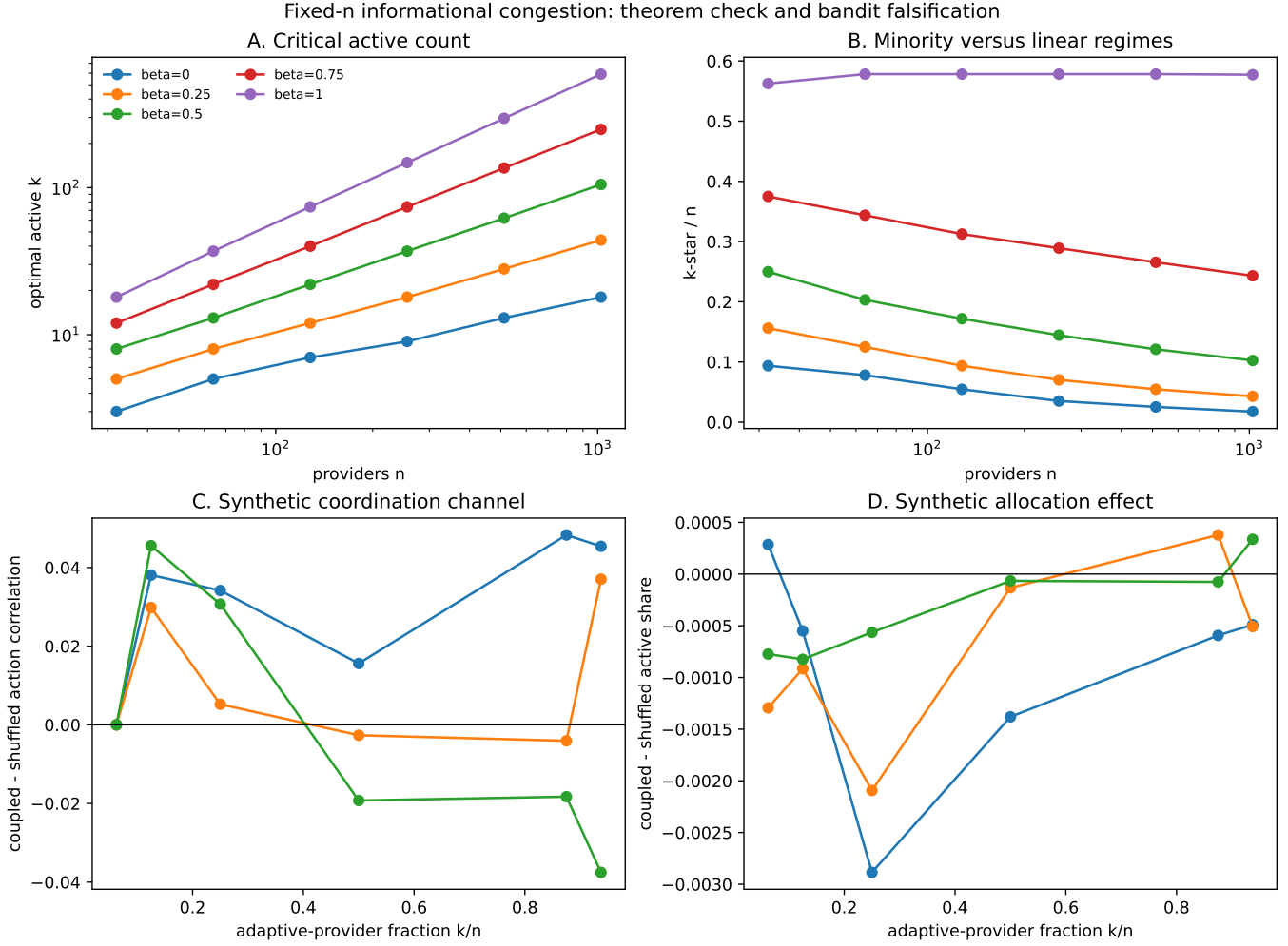


Figure 4: Conditional theorem check and negative control. Integer optima recover the predicted k^* exponent to within 0.006. The declared bandit intervention can move action correlation but has no distinguishable active-share effect; seed-clustered intervals include zero.

Proof of Theorem 1. Substituting (9) gives

$$V_n(k) = \frac{bk}{n} - \frac{ck^{2+\gamma}}{n^2 r_n^\gamma}.$$

For $k > 0$ the second derivative is strictly negative. The first-order condition is $b/n = c(2 + \gamma)k^{1+\gamma}/(n^2 r_n^\gamma)$, which yields (10). Strict concavity implies that the integer maximizer is one of the adjacent integers. Substituting $r_n \propto n^\beta$ gives (11); dividing by n proves the density statements. $\square$

Proof of Proposition 2. Substitution gives $(p_i^{\text{eff}})^{-a} = p_i^{-a} e^{\alpha_i}$, proving allocation equivalence. Taking the log odds of i and j yields $\log(s_i/s_j) = -a \log(p_i/p_j) + \alpha_i - \alpha_j$ and hence (14). Finally differentiate $\log s_i = -a \log p_i + h_i(\log p_i) - \log \sum_j p_j^{-a} e^{\alpha_j}$ with respect to $\log p_i$. The direct term is $-a + h'_i$ and its contribution to the log denominator is $s_i(-a + h'_i)$, giving $(1 - s_i)(-a + h'_i)$. $\square$

Proof of Theorem 2. The Gibbs minimizer satisfies $x_i^g = e^{-\eta g_i}/Z_g$, so $g_i = -\eta^{-1} \log x_i^g - \eta^{-1} \log Z_g$. Substitution into (19) gives

$$F_g(x) = \eta^{-1} \sum_i x_i \log \frac{x_i}{x_i^g} - \eta^{-1} \log Z_g.$$

The second term equals $F_g(x^g)$, proving (20). $\square$

Proof of Theorem 3. For every $x \in \Delta_n$, $|F_g(x) - F_{\hat{g}}(x)| = |\langle x, g - \hat{g} \rangle| \leq \epsilon$. Optimality of $x^{\hat{g}}$ for $F_{\hat{g}}$ therefore implies

$$F_g(x^{\hat{g}}) \leq F_{\hat{g}}(x^{\hat{g}}) + \epsilon \leq F_{\hat{g}}(x^g) + \epsilon \leq F_g(x^g) + 2\epsilon.$$

The lower bound follows from optimality of x^g . $\square$

Proof of Theorem 4. A quote-only router must use the same allocation in both observationally equivalent economies. Let x be the probability assigned to provider 1. Its regret is $\Delta(1 - x)$ in economy A and Δx in economy B; the larger is at least $\Delta/2$. If $x \in \{0, 1\}$, the larger is Δ . $\square$

B Evidence and Claim Ledger

The critical-set result is a theorem conditional on Assumption 1; SM3 validates its numerical scaling but its bandit allocation effect includes zero. WF20 uses 3,274 public snapshots and identifies a finite-menu rank diagnostic plus a deterministic shadow rule. It does not identify asymptotic rank, realized share, k^* , provider learning, or welfare. GLM-5.2 is linear-compatible, and the non-GLM census does not support linear-density active pricing.

The paid GLM panel identifies an owned-account effect of explicit price sorting in 15 complete blocks. It does not identify market-wide share, and its latent score, duration, and choice gates remain open. H91 and H92 establish a within– between difference and an accounting identity, not causal residual demand. WCV6 is a bounded sensitivity analysis. H81 identifies a narrow fallback decomposition, not a general router policy. Residual QoS tests and zero valid utilization cells do not identify technology rent, dumping, or marginal cost.

SM4 is a declared structural game. Its objective frontier, unilateral grid regret, technology stresses, and pairwise deviation gains are simulator objects. No result in this paper identifies router score memory, provider learning memory, communication, collusion, a provider algorithm, or live welfare.

C Reproducibility

The public manuscript panel is reproduced by scripts/run_paper_integrity_rerun.py. On the immutable revision reported above, all 29 registered modules complete with zero failures and no prospective owned outcomes queried. WF20 produces the rank and shadow-transfer figure. SM3 produces the fixed- n theorem check and signal-order negative control; SM4 produces the heterogeneous-provider objective frontier and technology stress panel. Unit tests cover rank algebra, fixed-provider menus, welfare accounting, capacity redistribution, objective selection, JSON serialization, and bounded best responses. The detailed claim audit is papers/information-congestion-evidence-audit-2026-07-22.md.

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